

Land & Site Due Diligence Report

Parcel control, physical constraints, access, easements, geography, and constructability.

Project: Solar Alpha **Location:** Solano County, California (fictional demo site) **Date:** August 14, 2026

Demo note: This document contains intentionally synthetic data and seeded inconsistencies for a one-day hackathon prototype. It is not a real engineering, legal, environmental, investment, or regulatory opinion.

Executive summary

Solar Alpha is a proposed 250 MW solar-plus-storage project across approximately 1,420 acres. The site is generally buildable, but title, access, drainage, and boundary conditions create material schedule and cost exposure.

Metric	Current finding	Risk signal
Site area	1,420 acres	Baseline
Controlled acreage	1,402 acres	18-acre gap
Slope > 8%	6.4% of site	Manageable
Mapped flood-prone area	12% of site	Design constraint
Permanent access points	2	One relies on disputed easement
Geotechnical borings	24 completed	2 show expansive clay

Parcel & title review

The project boundary includes 14 parcels. Thirteen have executed site-control agreements. Parcel 14, approximately 18 acres near the preferred substation route, is shown as project-controlled in the development schedule but the title summary lists the option agreement as unsigned.

RED FLAG - Parcel 14 is required for the current collector-line alignment. The development schedule assumes full site control by September 2026, but the executed documents do not support that assumption.

Easements and access

Primary construction access uses County Road 17. A secondary heavy-haul route crosses a 30-foot private easement. The title exhibit labels this easement "non-exclusive agricultural access" and does not explicitly permit oversize construction traffic.

WARNING - Heavy-haul access rights should be confirmed before notice to proceed; alternate routing adds an estimated 6.8 miles per delivery.

Geography and physical constraints

Constraint	Area affected	Potential impact
Seasonal drainage swales	9 corridors	Foundation setbacks / culverts
Flood-prone lowlands	~170 acres	Panel layout adjustment
Expansive clay	Northwest block	Road/base-course cost

Constraint	Area affected	Potential impact
High-wind exposure	Western ridge	Tracker design review
Existing irrigation lines	3 crossings	Relocation / owner coordination

Project Red Flag extraction targets

Recommended fields: controlled acreage %, unresolved title exceptions, easement sufficiency, flood exposure %, slope exposure %, geotechnical exceptions, access redundancy, land cost per acre, and land-condition precedent status.